

# Bryce Verberne

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## EDUCATION

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### B.S, Computer Science

Arizona State University, Tempe, AZ

Aug 2023 - Dec 2025

3.71 GPA

### A.S, Computer Science

Scottsdale Community College, Scottsdale, AZ

Aug 2021 - May 2023

3.91 GPA

## SKILLS

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**Programming Languages:** C, C++, Python, Java

**Tools & Technologies:** Git, GitHub, Jira, GDB, Docker, Linux, RTOS, Microcontrollers, MongoDB, Cameo Systems Modeler

**Engineering Practices:** Agile Methodologies, Unit Testing, Debugging, Software Design, Risk Management, Budgeting, V&V

## EXPERIENCE

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### Northrop Grumman: Software Engineering Intern

Jun 2025 – Present

- Supported NG's Missile Defense division by developing a multithreaded C++ pipeline streaming Cameo simulations into AFSIM.
- Developed MongoDB interface to track changes, map and validate jobs, and queue jobs for execution, storing results in SQLite.
- Performed unit, integration, stress, and regression tests to ensure system reliability and performance.
- Built HTML/CSS/JS GUI for real-time job tracking and authored SysML sequence/class diagrams in Cameo.
- Followed Agile workflows with Jira, using Git and Bitbucket under the OUTMATCH IRAD program.

### Sun Devil Satellite Laboratory Club

Apr 2024 – Present

#### ASCEND: Deputy Software Lead

- Co-led a 10-member software team using Scrum, Jira, and GitHub to design and develop flight software (C++), ground software (Python), and data processing tools for a NASA-funded high-altitude balloon project.
- Established weekly sprints and retrospectives, identified bottlenecks, and mentored team members to drive continuous advances.
- Refactored the legacy codebase to an OOP design in C++, streamlining new hardware integration for robust flight operations.
- Reduced data footprint by 67% by migrating from CSV-based logging to a packet-based system, improving storage efficiency.
- Enhanced SPI flash memory by designing onboard file system and payload commands for data management.
- Built a packet decoder for a multi-threaded Python ground software pipeline, enabling high-throughput real-time data processing.

#### Coconut CubeSat: Software Engineer

- Worked on a 7-member team to design and develop the software system for the Coconut CubeSat, a 1U satellite project.
- Wrote Python scripts to parse and route telemetry packages, facilitating satellite-to-ground communication.
- Integrated OpenC3 into ground software for hardline and radio communication using CCSDS protocols.

### NASA L'SPACE MCA

Jan 2024 – Apr 2024

#### Deputy Project Manager

- Co-led a team of 12 to design a conceptual Mars rover, overseeing milestone reviews including MCR, SRR, MDR, and PDR.
- Spearheaded programmatic sub-team, tracking budget, schedule, and risks.
- Facilitated monthly iterative phases with specific goals and deliverables, conducting regular reviews and incorporating feedback.
- Bridged Science and Engineering teams for effective cross-functional collaboration and risk management.

#### Computer Hardware Engineer

- Developed the Command and Data Handling (CDH) Subsystem of a conceptual Mars rover design, entailing the telecommunications, data computing, and software architecture subassemblies.
- Developed lower-level requirements and conducted trade studies to select sub-components.
- Identified risks, implemented redundancy, and determined mass, volume, and power specifications.
- Created software architecture flowchart and developed manufacturing and procurement plans.

## PERSONAL PROJECTS

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### SatComm Optimizer

May 2024 – Present

- Visualized real-time positions and capabilities of Iridium-NEXT satellites using CesiumJS and SGP4 libraries with TLE data.
- Integrated Python-driven data with JavaScript to depict satellite interactions with ground stations and inter-satellite links.
- Developing Python algorithms to optimize satellite communication processes, feeding performance data into visualizations.